

Finite Levels of the Weak-Compactness Index

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Status

This is a partial answer to Question 9.3 of Beanland–Causey–Freeman–Wallis, arXiv:1507.06285. The source asks for which ordinals ξ the class of operators $A : X \rightarrow Y$ satisfying

$$\mathbf{WC}(A, X, Y) \leq \xi$$

is an ideal. This packet settles the finite ordinal cutoffs.

Source Question

The source defines the weak-compactness index using the summing basis (s_i) of c_0 :

$$\mathbf{WC}(A, X, Y, K) = \{\emptyset\} \cup \{(x_i)_{i=1}^n \in B_X^{<\mathbb{N}} : (s_i)_{i=1}^n \lesssim_K (Ax_i)_{i=1}^n\},$$

and

$$\mathbf{WC}(A, X, Y) = \sup_{K \geq 1} o(\mathbf{WC}(A, X, Y, K)).$$

model indices. We have seen that a better estimate is possible for $\mathbf{NP}_\infty$, and for the $\mathbf{NP}_1$ index when the spaces involved have unconditional bases.

9.2. Weak compactness. Let (s_i) be the summing basis for c_0 . It follows from standard techniques modifying James's characterization of reflexivity that an operator $A : X \rightarrow Y$ fails to be weakly compact if and only if there exists $(x_i) \subset B_X$ so that (Ax_i) dominates (s_i) . Therefore for any operator $A : X \rightarrow Y$ and $K \geq 1$, we define

$$\mathbf{WC}(A, X, Y, K) = \{\emptyset\} \cup \left\{ (x_i)_{i=1}^n \in B_X^{<\mathbb{N}} : (s_i)_{i=1}^n \lesssim_K (Ax_i)_{i=1}^n \right\}.$$

We then let

$$\mathbf{WC}(A, X, Y) = \sup_{K \geq 1} o(\mathbf{WC}(A, X, Y, K)).$$

Then A is weakly compact if and only if $\mathbf{WC}(A, X, Y) < \infty$, which follows immediately from the definition. We make the following easy observations:

Proposition 9.1. (i) *The class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) \leq \omega$ is the ideal of super weakly compact operators.*

(ii) *The class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) < \omega_1$ is an ideal.*

The proof of part (i) is essentially the same as the proof that when (e_i) has property (S') , the class of operators $A : X \rightarrow Y$ so that $\mathbf{NP}_{(e_i)}(A, X, Y) \leq \omega$ is the ideal of all operators all ultrapowers of which fail to preserve a copy of (e_i) .

Of course, part (ii) would be trivial if we restricted our attention to separable domains, since if X is separable and $A : X \rightarrow Y$ is an operator, $\mathbf{WC}(A, X, Y)$ is either countable or ∞ . But since the $\mathbf{NP}_1$ index of an operator cannot be larger than the $\mathbf{WC}$ index, the identity operators on the reflexive examples W_ξ yield weakly compact operators having uncountable $\mathbf{WC}$ index. Thus part (ii) is non-trivial. Our proof of part (ii) follows by another descriptive set theoretic argument. Define the function $f : \mathfrak{L} \rightarrow \mathbf{Tr}$ by

$$(X, Y, \hat{A}) \mapsto \{\emptyset\} \cup \left\{ (k), k^\wedge (n_i)_{i=1}^l : (d_{n_i}(X))_{i=1}^l \in \mathbf{WC}(A, X, Y, k) \right\}.$$

Then by the same methods as in Lemma 8.5 $(X, Y, \hat{A}) \mapsto o(f(X, Y, \hat{A}))$ is a Π_1^1 rank on the Π_1^1 subset $\mathfrak{WC} \cap \mathfrak{L} = f^{-1}(\mathbf{WF})$ such that $o(f(X, Y, \hat{A})) = \mathbf{WC}(A, X, Y) + 1$. With this we establish the analogue of Theorem 8.2. There exists a function $\varphi_{WC} : [1, \omega_1) \times [1, \omega_1) \rightarrow [1, \omega_1)$ so that if $X, Y \in \mathbf{Ban}$ (not necessarily separable), $\xi, \zeta < \omega_1$ are such that $\mathbf{WC}(A, X, Y) \leq \xi$ and $\mathbf{WC}(B, X, Y) \leq \zeta$, then $\mathbf{WC}(A + B, X, Y) \leq \varphi_{WC}(\xi, \zeta)$. The proof is an inessential modification of the proof of Theorem 8.2.

Question 9.2. For which ordinals ξ is the class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) < \xi$

Question 9.3 is:

Π_1^1 subset $\mathfrak{WC} \cap \mathfrak{L} = f^{-1}(\mathbf{WF})$ such that $o(f(X, Y, A)) = \mathbf{WC}(A, X, Y) + 1$. With this we establish the analogue of Theorem 8.2. There exists a function $\varphi_{WC} : [1, \omega_1) \times [1, \omega_1) \rightarrow [1, \omega_1)$ so that if $X, Y \in \mathbf{Ban}$ (not necessarily seaparable), $\xi, \zeta < \omega_1$ are such that $\mathbf{WC}(A, X, Y) \leq \xi$ and $\mathbf{WC}(B, X, Y) \leq \zeta$, then $\mathbf{WC}(A + B, X, Y) \leq \varphi_{WC}(\xi, \zeta)$. The proof is an inessential modification of the proof of Theorem 8.2.

Question 9.3. *For which ordinals ξ is the class of operators $A : X \rightarrow Y$ such that $\mathbf{WC}(A, X, Y) \leq \xi$ an ideal?*

9.3. Other applications of $\mathfrak{L}$ and $\mathfrak{L}_1$. In some cases, it is perhaps more convenient to code only the operators having norm not exceeding 1. One convenience of $\mathfrak{L}_1$ is the following concerning the Szlenk index $Sz(A)$ of an operator. Recall that if X is a separable Banach space, $A : X \rightarrow Y$ is an Asplund operator if and only if $A^*B_{Y^*}$ is norm separable. Let $\mathfrak{A}$ denote the ideal of Asplund operators.

Theorem

Theorem. *Let $A : X \rightarrow Y$ be a bounded operator. For every finite integer $n \geq 1$,*

$$\mathbf{WC}(A, X, Y) \leq n \iff \text{rank}(A) \leq n - 1.$$

Equivalently, if $\text{rank}(A) = r < \infty$, then

$$\mathbf{WC}(A, X, Y) = r + 1,$$

while if A has infinite rank, then

$$\mathbf{WC}(A, X, Y) \geq \omega.$$

Proof Intuition

A finite node in the $\mathbf{WC}$ tree requires the image vectors to dominate the same number of summing-basis vectors. Domination of a basis forces linear independence of the image vectors. Thus a rank- r range cannot support a node of length $r + 1$. Conversely, any m independent vectors in the range of A can be lifted to $A(B_X)$ after scaling, and finite-dimensional basis equivalence gives some constant K for which they dominate the first m summing-basis vectors.

Proof

First suppose $\text{rank}(A) = r < \infty$. If $(x_i)_{i=1}^m \in \mathbf{WC}(A, X, Y, K)$, then $(Ax_i)_{i=1}^m$ K -dominates $(s_i)_{i=1}^m$. Since $(s_i)_{i=1}^m$ is linearly independent, the vectors $Ax_1, \dots, Ax_m$ must also be linearly independent: otherwise some nonzero scalar vector $(a_i)_{i=1}^m$ would satisfy

$$\sum_{i=1}^m a_i Ax_i = 0,$$

while

$$\left\| \sum_{i=1}^m a_i s_i \right\| > 0,$$

contradicting domination. Therefore every node has length at most r , and so

$$o(\mathbf{WC}(A, X, Y, K)) \leq r + 1$$

for every $K \geq 1$.

For the reverse inequality, choose linearly independent vectors $y_1, \dots, y_r$ in $A(B_X)$; if $r = 0$ there is nothing to choose. Pick $x_i \in B_X$ with $Ax_i = y_i$. Since $(y_i)_{i=1}^r$ and $(s_i)_{i=1}^r$ are bases of r -dimensional spaces, there exists $K \geq 1$ such that

$$(s_i)_{i=1}^r \lesssim_K (y_i)_{i=1}^r.$$

Thus $(x_i)_{i=1}^r$ is a node of $\mathbf{WC}(A, X, Y, K)$. A tree whose maximum node length is r and which has a node of length r has order $r + 1$. Hence $\mathbf{WC}(A, X, Y) = r + 1$.

If A has infinite rank, then for every m one can choose m linearly independent vectors in $A(B_X)$. The same finite-dimensional basis-equivalence argument gives a constant K_m such that $\mathbf{WC}(A, X, Y, K_m)$ has a node of length m . Consequently

$$\mathbf{WC}(A, X, Y) = \sup_{K \geq 1} o(\mathbf{WC}(A, X, Y, K)) \geq \sup_m (m + 1) = \omega.$$

This proves the theorem.

Consequence for Question 9.3

For finite $n \geq 1$,

$$\{A : X \rightarrow Y : \mathbf{WC}(A, X, Y) \leq n\} = \{A : X \rightarrow Y : \text{rank}(A) \leq n - 1\}.$$

For every $n \geq 2$, this class is not closed under sums. On an n -dimensional space, let P be a projection of rank $n - 1$ and Q the complementary rank-one projection. Then

$$\mathbf{WC}(P) \leq n, \quad \mathbf{WC}(Q) \leq n,$$

but

$$P + Q = I$$

has rank n , and hence $\mathbf{WC}(I) = n + 1 > n$.

For $n = 1$, the class is just the zero-operator class. It is closed under sums and composition, but it is not an operator ideal under conventions requiring all finite-rank operators to be included.

Verification Notes

The only finite-dimensional input is that any two bases of an m -dimensional normed space are equivalent, with a constant depending on the chosen bases. The proof permits this constant to depend on m , which is allowed because the definition of $\mathbf{WC}(A, X, Y)$ takes the supremum over all $K \geq 1$.

Search Bounds

The local registry, solution index, attempt index, and proof-gap index were searched for the source arXiv id, weak compactness index, $\mathbf{WC}(A, X, Y)$, finite rank, summing basis, and operator-index terminology. Web searches for the source title and arXiv id together with $\mathbf{WC}(A, X, Y)$, finite rank, and ideal phrasing found no direct later answer to this finite-cutoff observation.

Limitations

This packet only settles finite ordinal cutoffs in Question 9.3. It does not classify infinite countable ordinals. The source already proves that the cutoff $\mathbf{WC} \leq \omega$ is the ideal of super weakly compact operators and that $\mathbf{WC} < \omega_1$ is an ideal.

Reference

K. Beanland, R. Causey, D. Freeman, B. Wallis, *Classes of operators determined by ordinal indices*, arXiv:1507.06285.